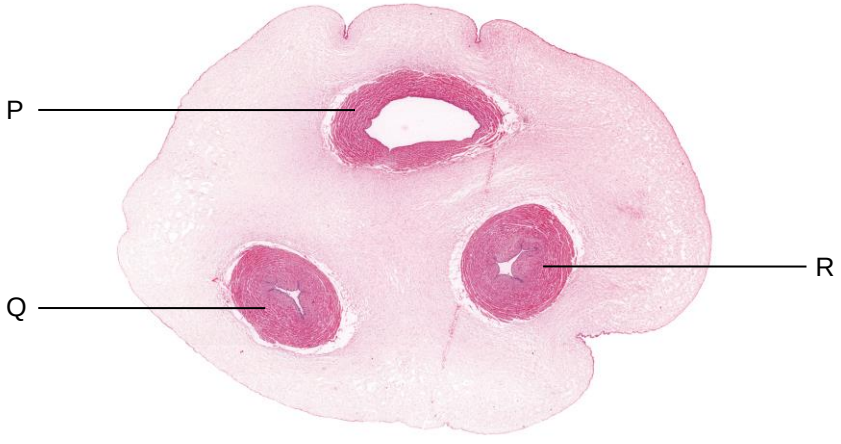
Multiple-choice questions

|!|BCEMB0140000001|!| ★ **Cross topic**

Directions: Questions (a) and (b) refer to the photomicrograph below, which shows a section of a human umbilical cord with three blood vessels.



(a) Which of the following blood vessels contain(s) deoxygenated blood?

A. P only
B. Q only
C. P and R only
D. Q and R only

##D##

Skill assessed	Mark
Analysing data	1

|!|BCEMB0140000002|!| ★ **Cross topic**

(b) Which of the following descriptions of the blood vessels is/are correct?

(1) The blood pressure in vessel Q is higher than that in vessel P.
(2) The blood in vessel Q has a higher amino acid content than that in vessel R.
(3) The blood in vessel R has a higher urea content than that in vessel P.

A. (1) only
B. (2) only
C. (1) and (3) only
D. (2) and (3) only

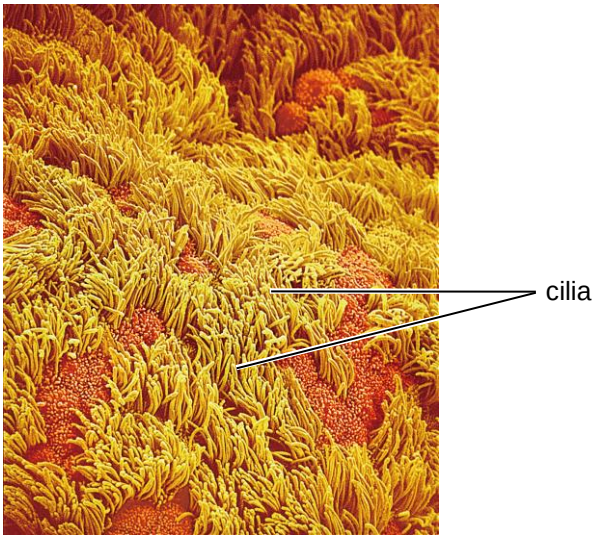
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##C##

Skill assessed	Mark
Analysing data	1

|!|BCEMB0140000003|!| ★ **Cross topic**

The photomicrograph below shows a type of epithelium in the human body.



This type of epithelium can be found in the

(1) nasal cavity.
(2) oviduct.
(3) bronchus.

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

□

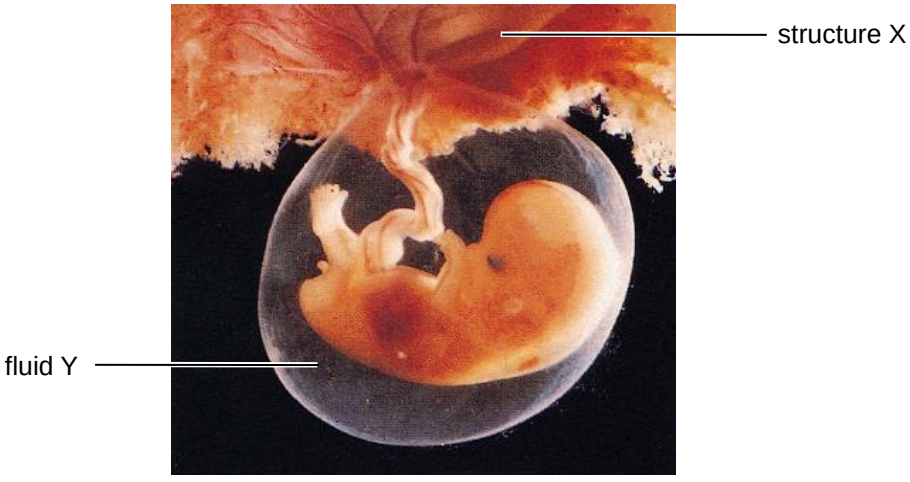
##D##

Skill assessed	Mark
Analysing data	1

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|!|BCEMB0140000004|!| ★

Directions: Questions (a) to (c) refer to the photograph below, which shows a foetus and the associated structures inside the uterus.



(a) Which of the following statements about structure X are correct?

(1) It is rich in blood supply.
(2) It protects the foetus from mechanical shock.
(3) It is composed of tissues from both the mother and the foetus.

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

##B##

Skill assessed	Mark
Analysing data	1

|!|BCEMB0140000005|!| ★ **Cross topic**

(b) Which of the following structures of a mung bean seed has a similar function to fluid Y?

A. radicle
B. cotyledon
C. seed coat
D. micropyle

##C##

Skill assessed	Mark
Analysing data	1

|!|BCEMB0140000006|!| ★

(c) Which of the following can pass from the maternal blood to the foetal blood through structure X?

(1) viruses
(2) antibodies
(3) nicotine

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

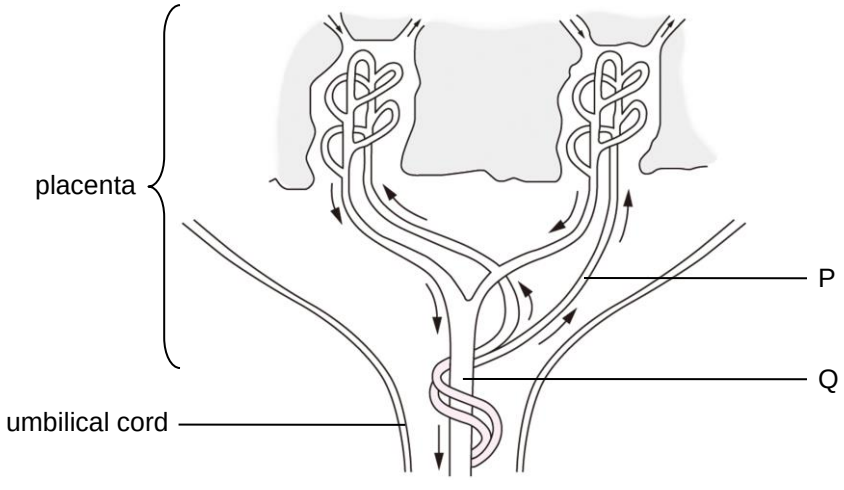
□

##D##

Skill assessed	Mark
Analysing data	1

|!|BCEMB0140000007|!| ★

The diagram below shows part of the placenta and the umbilical cord with blood vessels in humans.



Which of the following comparisons of blood vessels P and Q is/are correct?

(1) The blood pressure in vessels P and Q is the same.
(2) The blood in vessel P has a higher oxygen content than that in vessel Q.
(3) The blood in vessel P has a higher urea content than that in vessel Q.

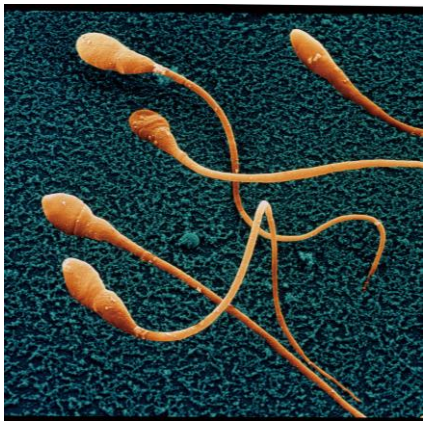
A. (3) only
B. (1) and (2) only
C. (1) and (3) only
D. (2) and (3) only

##A##

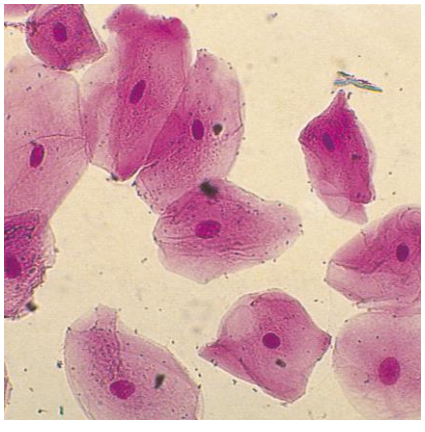
Skill assessed	Mark
Analysing data	1

|!|BCEMB0140000008|!| ★

The photomicrographs below show two types of cells in humans.



Cell type P



Cell type Q

Which of the following correctly compare cell types P and Q?

(1) Both contain mitochondria.
(2) Both can carry out mitotic cell division.
(3) Cell type P contains 23 chromosomes while cell type Q contains 46 chromosomes.

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

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##B##

Skill assessed	Mark
Analysing data	1

|!|BCEMB0140000009|!| ★ **Cross topic**

Which of the following correctly compare an ovum and a mature red blood cell in humans?

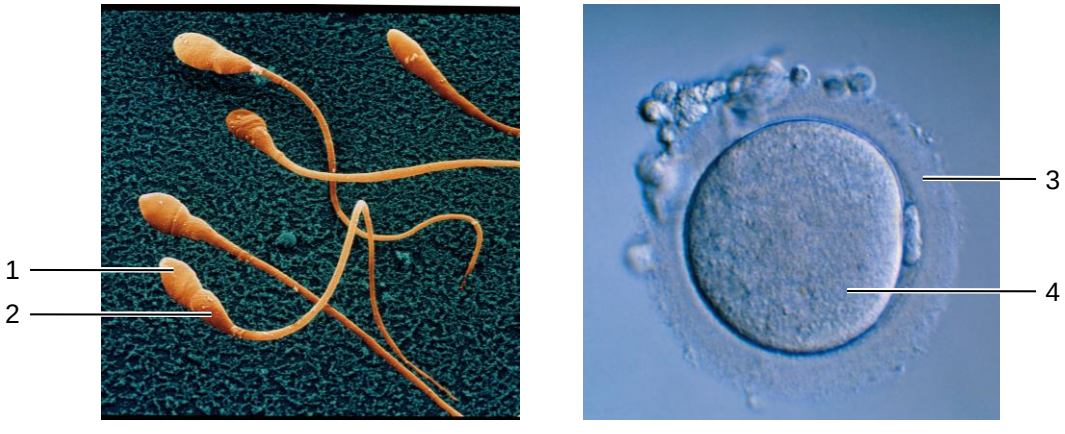
A. Both contain mitochondria.
B. Both have the same amount of cytoplasm.
C. Both have the same amount of DNA.
D. Both cannot carry out mitosis.

##D##

Skill assessed	Mark
Applying concepts	1

|!|BCEMB0140000010|!| ★

The photomicrographs below show some human sperms and a human ovum.



Human sperm ***Human ovum***

Which of the following structures provides nutrients for the development of the embryo in the first 5 days?

A. 1
B. 2
C. 3
D. 4

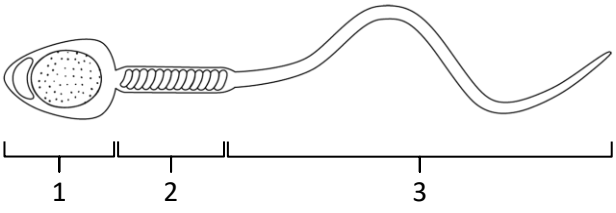
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##D##

Skill assessed	Mark
Analysing data	1

|!|BCEMB0140000011|!| ★

The diagram below shows the structure of a human sperm.



Which of the following will happen when the sperm meets an ovum in the oviduct of the female body?

(1) Part 1 of the sperm releases enzymes to break down the jelly coat of the ovum.
(2) During fertilization, part 3 of the sperm will be left outside the ovum.
(3) Part 2 of the sperm will contribute mitochondria to the zygote formed.

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

##A##

Skill assessed	Mark
Analysing data	1

|!|BCEMB0140000012|!| ★

The photomicrograph below shows a four-day-old human embryo.



Which of the following statements about the embryo is/are correct?

(1) The embryo is undergoing cell division and cell differentiation.
(2) The embryo is ready for implantation.
(3) The embryo can be moved along the oviduct by the beating action of the cilia.

A. (3) only
B. (1) and (2) only
C. (1) and (3) only
D. (2) and (3) only

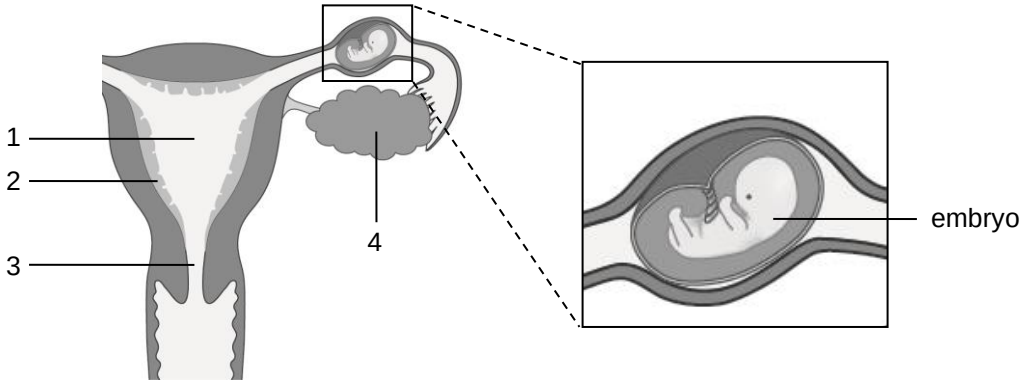
□

##A##

Skill assessed	Mark
Analysing data	1

|!|BCEMB0140000013|!| ★ **STSE**

Directions: Questions (a) and (b) refer to the diagram below, which shows a type of abnormal pregnancy in which the embryo develops in the oviduct.



(a) In which of the following structures does implantation normally take place?

A. 1
B. 2
C. 3
D. 4

##B##

Skill assessed	Mark
Analysing data	1

|!|BCEMB0140000014|!| ★ **STSE**

(b) Which of the following are dangers caused by this type of abnormal pregnancy to the mother?

(1) The oviduct may rupture as the embryo grows bigger.
(2) Severe pain may occur on the abdomen.
(3) Excessive bleeding may occur.

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

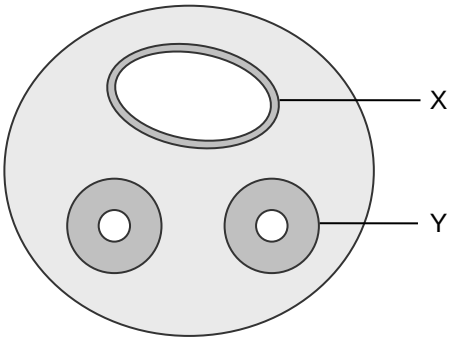
##D##

Skill assessed	Mark
Analysing data	1

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|!|BCEMB0140000015|!| ★

The diagram below shows a section of a human umbilical cord with blood vessels.



Which of the following combinations correctly shows the directions of blood flow in vessels X and Y?

	X	***Y***
A.	towards the foetus	towards the mother
B.	towards the foetus	away from the mother
C.	away from the foetus	towards the mother
D.	away from the foetus	away from the mother

##A##

Skill assessed	Mark
Analysing data	1

|!|BCEMB0140000016|!| ★

During the development of foetus in the uterus, the red blood cells of the foetus obtain oxygen from the red blood cells of the mother. Which of the following statements correctly explain this?

A. The foetal red blood cells contain more haemoglobin than the maternal red blood cells.
B. The foetal red blood cells are larger in size than the maternal red blood cells.
C. Foetal haemoglobin has a higher affinity for oxygen than maternal haemoglobin.
D. The number of red blood cells per unit volume in foetal blood is larger than that of the maternal blood.

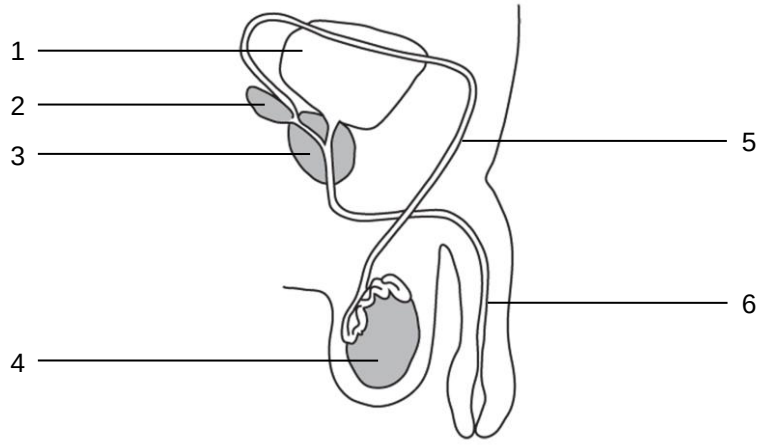
##C##

Skill assessed	Mark
Applying concepts	1

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|!|BCEMC0140000017|!|

Directions: Questions (a) to (c) refer to the diagram below, which shows the reproductive system of a man.



(a) The components of semen are produced by

A. 1 and 2 only.
B. 2 and 3 only.
C. 3 and 4 only.
D. 2, 3 and 4 only.

##D##

Skill assessed	Mark
Analysing data	1

|!|BCEMC0140000018|!|

(b) Which of the following combinations correctly shows the functions of structures 1 and 5?

	Structure 1	***Structure 5***
A.	produces urine	transfers semen
B.	produces urine	transfers sperms
C.	stores urine	transfers sperms
D.	stores urine	transfers semen

##C##

Skill assessed	Mark
Analysing data	1

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|!|BCEMB0140000019|!| ★

(c) Which of the following may still occur after structure 5 on both sides of the body has been tied and cut?

(1) production of sex hormones
(2) production of sperms
(3) ejaculation

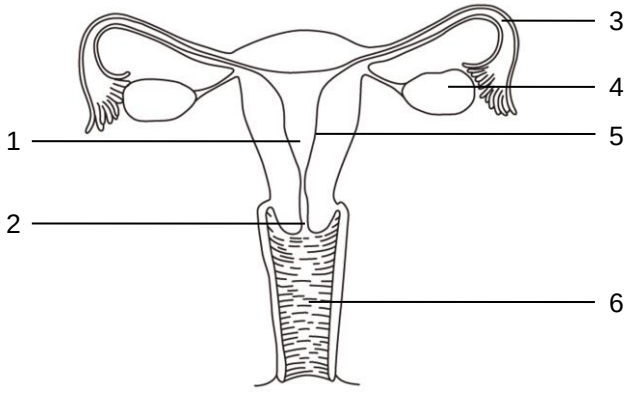
A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

##D##

Skill assessed	Mark
Analysing data	1

|!|BCEMC0140000020|!|

Directions: Questions (a) to (d) refer to the diagram below, which shows the female reproductive system.



(a) Which of the following combinations correctly identifies structures 2 and 5?

	Structure 2	***Structure 5***
A.	vagina	uterus
B.	cervix	uterus
C.	vagina	uterine lining
D.	cervix	uterine lining

##D##

Skill assessed	Mark
Analysing data	1

|!|BCEMC0140000021|!|

(b) Fertilization usually takes place in structure

A. 1.

B. 3.

C. 4.

D. 6.

##B##

Skill assessed	Mark
Analysing data	1

|!|BCEMC0140000022|!|

(c) Implantation usually taken place in structure

A. 1.

B. 2.

C. 3.

D. 5.

##D##

Skill assessed	Mark
Analysing data	1

|!|BCEMC0140000023|!|

(d) A woman is going to use intrauterine device for contraception. Where should it be placed?

A. 1

B. 2

C. 3

D. 6

##A##

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Skill assessed	Mark
Analysing data	1

|!|BCEMC0140000024|!|

In a mature sperm, numerous mitochondria are found

A. in the head.
B. in the midpiece.
C. in the tail.
D. evenly spaced throughout the cell.

##B##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0140000025|!|

Which of the following contraceptive methods prevent sperms from reaching an ovum?

(1) using a condom
(2) using an intrauterine device
(3) using the rhythm method

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

##B##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0140000026|!|

Which of the following statements about an intrauterine device is correct?

A. It should be renewed every month.
B. It prevents fertilization to take place.
C. It kills both the ovum and sperms.
D. It prevents implantation to take place.

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##D##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0140000027|!|

Which of the following ***cannot*** be passed from the mother to the foetus through the placenta?

A. alcohol
B. antibodies
C. oxygen
D. red blood cells

##D##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0140000028|!|

Which of the following comparisons about the compositions of blood in the umbilical artery and the umbilical vein is correct?

	Blood in the umbilical artery	***Blood in the umbilical vein***
A.	more glucose	less glucose
B.	less urea	more urea
C.	less oxygen	more oxygen
D.	less carbon dioxide	more carbon dioxide

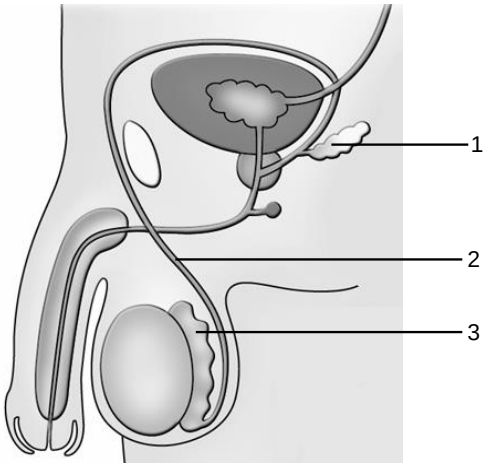
##C##

Skill assessed	Mark
Understanding basic concepts	1

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|!|BCEMC0140000029|!|

The following diagram shows the human male reproductive system.



Which of the following combinations correctly matches the name of each labelled structure?

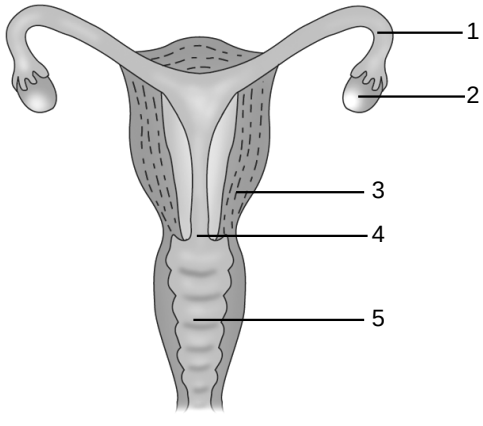
	Structure 1	***Structure 2***	***Structure 3***
A.	seminal vesicle	ureter	epididymis
B.	prostate gland	ureter	testis
C.	seminal vesicle	sperm duct	epididymis
D.	prostate gland	sperm duct	testis

##C##

Skill assessed	Mark
Analysing data	1

|!|BCEMC0140000030|!|

Directions: Questions (a) to (c) refer to the diagram below which shows the human female reproductive system.



(a) Which of the following combinations correctly identifies structures 1, 3 and 5?

	Structure 1	*Structure 3*	*Structure 5*
A.	oviduct	uterus	vagina
B.	oviduct	vagina	uterus
C.	vagina	uterus	oviduct
D.	vagina	oviduct	uterus

##A##

Skill assessed	Mark
Analysing data	1

|!|BCEMC0140000031|!|

(b) Which of the following statements about structure 2 is/are correct?

(1) Fertilization usually takes place in structure 2.
(2) Oestrogen is released from structure 2.
(3) An ovum is usually released from structure 2 each month.

A. (3) only
B. (1) and (2) only
C. (1) and (3) only
D. (2) and (3) only

##D##

Skill assessed	Mark
Analysing data	1

|!|BCEMC0140000032|!|

(c) Contraception can be achieved by using a diaphragm to cover structure

A. 1.
B. 3.
C. 4.
D. 5.

##C##

Skill assessed	Mark
Analysing data	1

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|!|BCEMB0140000033|!| ★

Which of the following structures in the male reproductive system is an endocrine gland?

A. prostate gland
B. testis
C. Cowper's gland
D. seminal vesicle

##B##

Skill assessed	Mark
Applying concepts	1

|!|BCEMC0140000034|!|

Which of the following statements about the male reproductive system is ***incorrect***?

A. The testes secrete the seminal fluid.
B. The epididymis is responsible for temporary storage of sperms.
C. The sperm ducts connect to the urethra.
D. The prostate gland secretes fluid that can nourish the sperms.

##A##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0140000035|!|

Where does fertilization take place in females?

A. ovary
B. vagina
C. uterus
D. oviduct

##D##

Skill assessed	Mark
Understanding basic concepts	1

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|!|BCEMC0140000036|!|

Which of the following hormones is/are produced by the yellow body?

(1) oestrogen
(2) insulin
(3) progesterone

A. (1) only
B. (3) only
C. (1) and (2) only
D. (1) and (3) only

##D##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0140000037|!|

The implantation of an embryo normally occurs in

A. the oviduct.
B. the uterus.
C. the vagina.
D. the ovary.

##B##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0140000038|!|

Which of the following is/are the function(s) of amniotic fluid?

(1) providing nutrients to the embryo
(2) acting as a shock absorber
(3) moistening the developing embryo

A. (3) only
B. (1) and (2) only
C. (1) and (3) only
D. (2) and (3) only

##D##

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Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMB0140000039|!| ★

Which of the following descriptions about the human umbilical vein is ***incorrect***?

A. The umbilical vein carries deoxygenated blood.
B. The umbilical vein contains valves.
C. The umbilical vein carries blood towards the embryo.
D. There is only one umbilical vein in the umbilical cord.

□

##A##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0140000040|!|

Which of the following substances can pass through the placenta and enter the foetal blood?

(1) antibodies
(2) red blood cells
(3) amino acids

A. (3) only
B. (1) and (2) only
C. (1) and (3) only
D. (2) and (3) only

□

##C##

Skill assessed	Mark
Understanding basic concepts	1

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|!|BCEMC0140000041|!|

Which of the following statements about the contraceptive methods is ***incorrect***?

A. Intrauterine devices can prevent ovulation.

B. Diaphragms can prevent sperms from reaching the ovum.

C. After vasectomy, the semen ejaculated does not contain sperms.

D. The use of condoms can prevent the spread of gonorrhoea.

##A##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0140000042|!|

Which of the following contraceptive method has the ***least*** side effects?

A. rhythm method

B. applying a contraceptive patch to the skin

C. taking oral contraceptive pills

D. using intrauterine devices

##A##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0140000043|!|

Which of the following statements about the man who has performed vasectomy is ***incorrect***?

A. Male sex hormones can still be produced.

B. Sperms can still be produced.

C. The semen contains no sperms.

D. No ejaculation can occur.

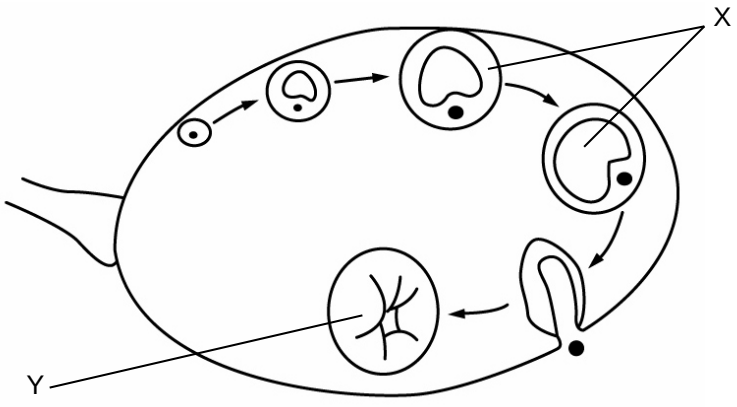
##D##

Skill assessed	Mark
Understanding basic concepts	1

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|!|BCEMC0140000044|!|

The figure below shows part of the female reproductive system.



What of the following combinations correctly identifies structures X and Y?

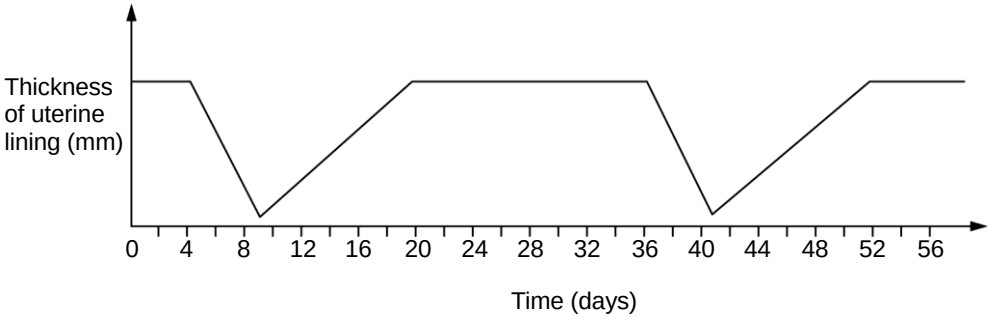
	Structure X	***Structure Y***
A.	ovary	ovum
B.	ovum	yellow body
C.	follicle	yellow body
D.	follicle	ovary

##C##

Skill assessed	Mark
Analysing data	1

|!|BCEMB0140000045|!| ★

Directions: Questions (a) to (c) refer to the graph below which shows the changes in the thickness of the uterine lining in two months.



(a) When does menstruation occur?

A. day 0-4

B. day 4-8

C. day 8-20

D. day 20-36

##B##

Skill assessed	Mark
Analysing data	1

|!|BCEMB0140000046|!| ★

(b) Ovulation occurs on

A. day 0 and day 35.

B. day 4 and day 36.

C. day 9 and day 41.

D. day 20 and day 52.

##D##

Skill assessed	Mark
Analysing data	1

|!|BCEMB0140000047|!| ★

(c) How long does this menstrual cycle last?

A. 5 days

B. 16 days

C. 28 days

D. 32 days

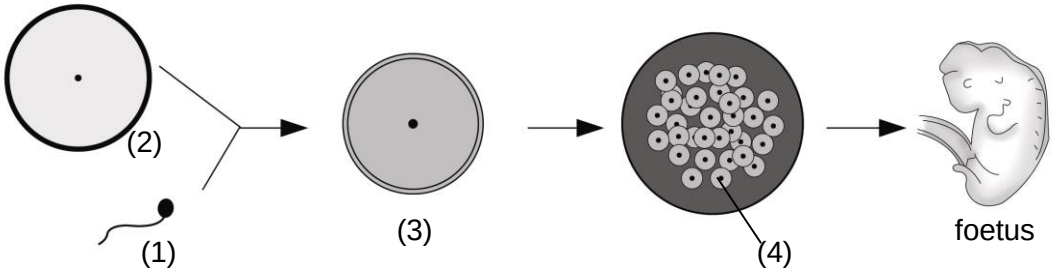
##D##

Skill assessed	Mark
Analysing data	1

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|!|BCEMB0140000048|!| ★

Directions: Questions (a) and (b) refer to the following diagram which shows the stages of the formation of a human foetus.



(a) Which of the above cells are diploid?

A. (1) and (2) only
B. (2) and (3) only
C. (3) and (4) only
D. (2), (3) and (4) only

☐

##C##

Skill assessed	Mark
Analysing data	1

|!|BCEMB0140000049|!| ★

(b) The foetus is surrounded by a fluid. Which of the following is ***not*** a function of the fluid?

A. It acts as a shock absorber.
B. It lubricates the vagina during labour.
C. It supplies nutrients to the foetus.
D. It allows the foetus some freedom of movement.

☐

##C##

Skill assessed	Mark
Understanding basic concepts	1

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|!|BCEMC0140000050|!|

Which of the following substances can be transmitted from the maternal blood to the foetal blood?

(1) nicotine
(2) glucose
(3) antibodies

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

##D##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0140000051|!|

Which of the following statements about twins are correct?

(1) Identical twins are genetically identical.
(2) Identical twins may have different heights.
(3) Fraternal twins are always of the same sexes.

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

##A##

Skill assessed	Mark
Understanding basic concepts	1

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|!|BCEMC0140000052|!|

Which of the following are the reasons to explain why only a small portion of sperms can reach the oviduct after the sperms are discharged into the vagina?

(1) Some sperms may be trapped by the mucus in the vagina.
(2) The acidic environment of the vagina is not favourable for the survival of sperms.
(3) Some sperms are destroyed by the white blood cells.

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

##A##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0140000053|!|

Implantation of embryo normally occurs in the

A. oviduct.
B. uterine lining.
C. cervix.
D. vagina.

##B##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0140000054|!|

Which of the following functions are carried out by the testes in humans?

(1) production of sperms
(2) production of male sex hormones
(3) secretion of seminal fluid

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

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##A##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0140000055|!|

Which of the following combinations about the structures of male reproductive system is ***incorrect***?

	Structure	***Function***
A.	penis	copulation
B.	prostate glands	secretion of seminal fluid
C.	epididymis	temporary storage of sperms
D.	scrotum	production of sperms

##D##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0140000056|!|

Semen includes

(1) sperms.
(2) sugars.
(3) mucus.

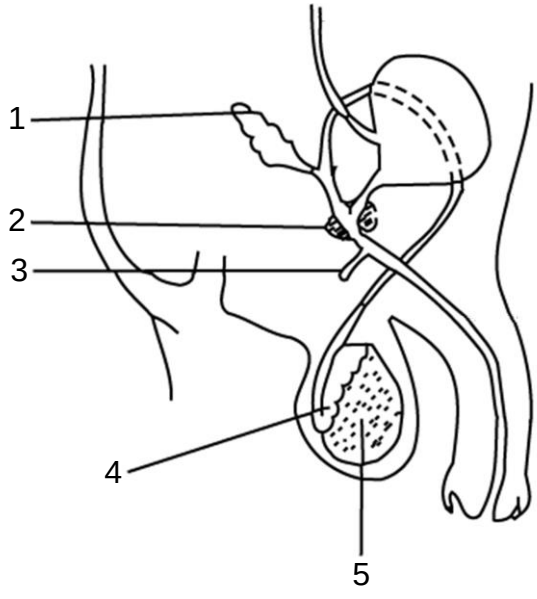
A. (1) only
B. (1) and (2) only
C. (2) and (3) only
D. (1), (2) and (3)

##D##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0140000057|!|

Directions: Questions (a) to (c) refer to the diagram below which shows the side view of the human male reproductive system.



(a) The function(s) of structure 2 is/are to

(1) produce sperms.
(2) secrete a fluid which nourishes the sperms.
(3) secrete a fluid to neutralize the acidic secretions of the female's vagina.

A. (2) only
B. (1) and (2) only
C. (1) and (3) only
D. (2) and (3) only

##D##

Skill assessed	Mark
Analysing data	1

|!|BCEMC0140000058|!|

(b) Sperms are temporarily stored in structure

A. 1.

B. 3.

C. 4.

D. 5.

##C##

Skill assessed	Mark
Analysing data	1

|!|BCEMC0140000059|!|

(c) Which of the following statements about structure 5 are correct?

(1) It is the site where meiosis occurs in males.

(2) It produces male sex hormones.

(3) It contains numerous sperm ducts.

A. (1) and (2) only

B. (1) and (3) only

C. (2) and (3) only

D. (1), (2) and (3)

##A##

Skill assessed	Mark
Analysing data	1

|!|BCEMC0140000060|!|

Which of the following would occur if the sperm duct were cut in a man?

A. The man would be unable to urinate.

B. The man would become sterile.

C. The man would be unable to produce sperms.

D. The man would be unable to produce male sex hormones.

##B##

Skill assessed	Mark
Understanding basic concepts	1

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|!|BCEMC0140000061|!|

Which of the following statements about the structures of female reproductive system is ***incorrect***?

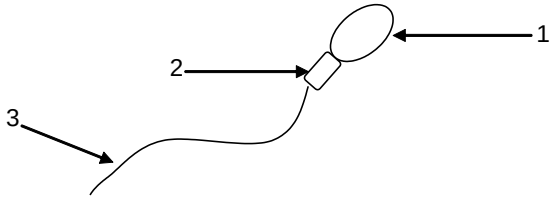
A. The urethra has both reproductive and excretory functions.
B. The ovary is responsible for producing ova and female sex hormones.
C. Fertilization usually takes place in the oviduct.
D. The uterus is the site of foetal development.

##A##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMB0140000062|!| ★

The following diagram shows a human male gamete.



Which of the following statements about this gamete are ***correct***?

(1) Structure 3 can help the gamete swim in the female reproductive tract.
(2) Structure 2 is packed with mitochondria.
(3) Structure 1 does not contain maternal chromosomes.

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

##A##

Skill assessed	Mark
Analysing data	1

|!|BCEMB0140000063|!| ★ **Cross topic**

Which of the following statements about ova is ***incorrect***?

A. Ova are not motile.

B. Ova have food reserves.

C. The jelly coat of ova thickens and hardens after fertilization.

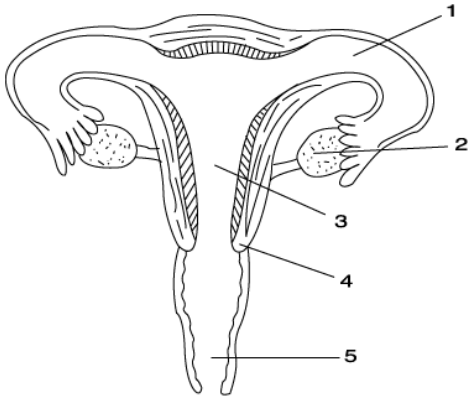
D. The chromosomes in the ova determine the sex of the offspring.

##D##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0140000064|!|

Directions: Questions (a) and (b) refer to the diagram below which shows the human female reproductive system.



(a) The hormone(s) produced in structure 2 include(s)

(1) oestrogen.
(2) glucagon.
(3) progesterone.

A. (1) only

B. (1) and (3) only

C. (2) and (3) only

D. (1), (2) and (3)

##B##

Skill assessed	Mark
Analysing data	1

|!|BCEMC0140000065|!|

(b) A diaphragm can be used for contraception. It is placed inside the female reproductive system and covers structure

A. 1.

B. 3.

C. 4.

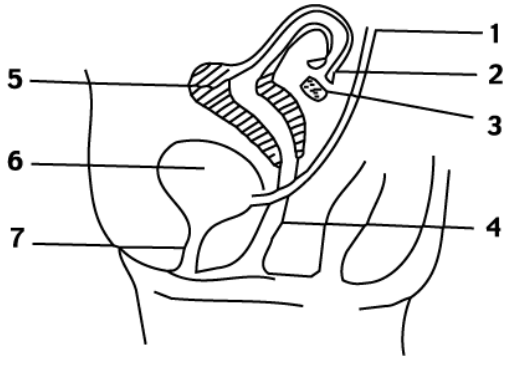
D. 5.

##C##

Skill assessed	Mark
Analysing data	1

|!|BCEMC0140000066|!|

The following diagram shows the female reproductive system.



Which of the following structures are ***not*** involved in reproduction?

A. 1, 4 and 6 only

B. 2, 3 and 4 only

C. 1, 6 and 7 only

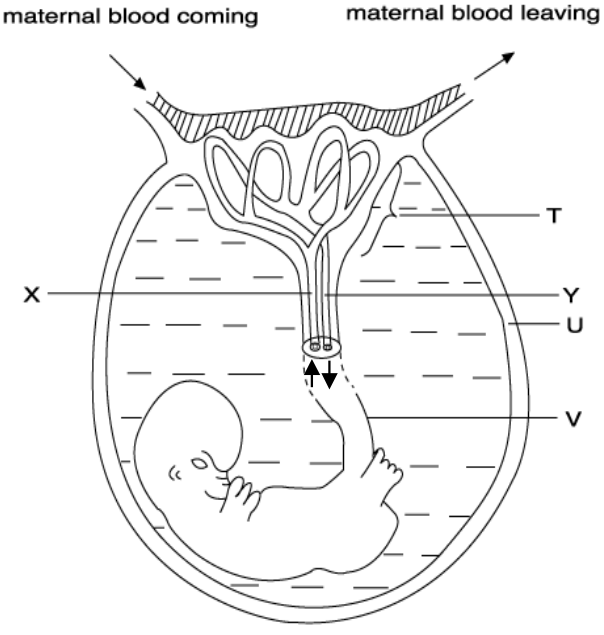
D. 2, 5 and 6 only

##C##

Skill assessed	Mark
Analysing data	1

|!|BCEMB0140000067|!| ★

Directions: Questions (a) to (c) refer to the diagram below which shows a developing embryo inside the uterus of a woman.



(a) Which of the following is/are the function(s) of structure T?

(1) allows the exchange of materials between mother and embryo
(2) secretes hormones to maintain the thickness of uterine lining
(3) acts as a shock-absorber

A. (1) only
B. (1) and (2) only
C. (2) and (3) only
D. (1), (2) and (3)

##B##

Skill assessed	Mark
Analysing data	1

|!|BCEMB0140000068|!| ★

(b) Which of the following comparisons between the blood in vessels X and Y is ***incorrect***?

	Blood in vessel X	***Blood in vessel Y***
A.	oxygenated	deoxygenated
B.	higher blood pressure	lower blood pressure
C.	more urea	less urea
D.	less glucose	more glucose

##A##

Skill assessed	Mark
Analysing data	1

|!|BCEMB0140000069|!| ★

(c) Which of the following is ***not*** a function of the fluid inside U?

A. protects the embryo from mechanical shock
B. acts as a medium through which gases can be exchanged between mother and embryo
C. provides an environment with constant temperature
D. allows the embryo to have some freedom of movement

##B##

Skill assessed	Mark
Analysing data	1

|!|BCEMC0140000070|!|

Gestation period is

A. the time period in which an ovum is fertilized by a sperm after copulation.
B. the time period in which a zygote develops into a foetus that is ready to be born.
C. the time period in which the mother gives birth to a baby.
D. the time period in which a follicle matures completely.

##B##

Skill assessed	Mark
Understanding basic concepts	1

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|!|BCEMC0140000071|!|

Which of the following comparisons between male and female gametes in humans is ***incorrect***?

	Male gamete	***Female gamete***
A.	motile	non-motile
B.	produced in large numbers every day	usually one is released each month
C.	with a smaller amount of food reserves	with a larger amount of food reserves
D.	contains mitochondria	does not contain mitochondria

##D##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0140000072|!|

Which of the following statements about ovulation is/are correct?

(1) It involves the bursting of a yellow body.
(2) It usually takes place once a month in humans.
(3) It is usually followed by the thickening of the uterine lining.

A. (3) only
B. (1) and (2) only
C. (2) and (3) only
D. (1), (2) and (3)

##C##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0140000073|!|

Which of the following statements about fraternal twins is/are correct?

(1) They come from different zygotes.
(2) They are genetically different.
(3) They are always of the same sex.

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

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##A##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0140000074|!|

Which of the following cells is/are diploid?

(1) ovum
(2) zygote
(3) sperm producing cells

A. (1) only
B. (2) only
C. (1) and (2) only
D. (2) and (3) only

##D##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0140000075|!|

Which of the following statements about a human zygote is/are correct?

(1) It is formed by the fertilization of an ovum by a sperm.
(2) It divides by meiotic cell division to develop into an embryo.
(3) It is diploid.

A. (3) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

##B##

Skill assessed	Mark
Understanding basic concepts	1

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|!|BCEMC0140000076|!|

Which of the following are the benefits of breast feeding?

(1) It supplies the right amount of all the essential food substances to the baby.
(2) The protein in the breast milk is easy for babies to digest and absorb.
(3) It reduces the risk of ovarian cancer for the mother.

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

##D##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0140000077|!|

Which of the following is/are the adaptation(s) of the placenta?

(1) It is composed of numerous villi.
(2) It allows the mixing of the embryo's blood with the maternal blood.
(3) It is connected to the umbilical cord.

A. (3) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

##B##

Skill assessed	Mark
Understanding basic concepts	1

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|!|BCEMC0140000078|!|

Which of the following statements about a pregnant woman are correct?

(1) Mammary glands of a pregnant woman develop and become ready to produce milk.
(2) No menstruation occurs in a pregnant woman.
(3) No ovulation occurs in a pregnant woman.

A. (1) only
B. (1) and (2) only
C. (2) and (3) only
D. (1), (2) and (3)

##D##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0140000079|!|

Which of the following shows the correct path of a sperm after it is ejaculated into the female body?

A. cervix → vagina → uterus → oviduct
B. cervix → vagina → oviduct → uterus
C. vagina → cervix → uterus → oviduct
D. vagina → uterus → cervix → oviduct

##C##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0140000080|!|

Which of the following indicates the correct path of a sperm before it is ejaculated into a woman's body?

A. scrotum → testis → epididymis → ureter
B. scrotum → testis → epididymis → urethra
C. testis → epididymis → sperm duct → ureter
D. testis → epididymis → sperm duct → urethra

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##D##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0140000081|!|

The following are the processes that take place in a female during reproduction.

(1) fertilization
(2) labour
(3) copulation
(4) implantation

Which of the following is the correct order of the above processes?

A. (1) → (2) → (3) → (4)
B. (1) → (3) → (4) → (2)
C. (3) → (1) → (4) → (2)
D. (3) → (4) → (1) → (2)

##C##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMB0140000082|!| ★

Which of the following statements about the function(s) of different contraceptive methods is/are correct?

(1) Oral contraceptive pills contain synthetic sex hormones which prevent ovulation.
(2) Vasectomy prevents pregnancy by terminating the production of sperms.
(3) A diaphragm can prevent the implantation of an embryo.

A. (1) only
B. (1) and (2) only
C. (2) and (3) only
D. (1), (2) and (3)

##A##

Skill assessed	Mark
Understanding basic concepts	1

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|!|BCEMB0140000083|!| ★ **Cross topic**

Which of the following statements about the oxygen content in different blood vessels are correct?

(1) The umbilical vein carries deoxygenated blood.
(2) The renal artery carries oxygenated blood.
(3) The pulmonary artery carries deoxygenated blood.

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

##C##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMB0140000084|!| ★

Which of the following functions can the placenta serve for the foetus?

(1) gas exchange
(2) excretory organ
(3) endocrine organ

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

##D##

Skill assessed	Mark
Applying concepts	1

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|!|BCEMC0140000085|!|

Which of the following comparisons about the compositions of blood in the umbilical artery and the umbilical vein is ***incorrect***?

	Umbilical artery	***Umbilical vein***
A.	less glucose	more glucose
B.	more urea	less urea
C.	less oxygen	more oxygen
D.	less carbon dioxide	more carbon dioxide

##D##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0140000086|!|

Which of the following statements about the placenta is correct?

(1) It is composed of many villi.
(2) It is formed from the tissues of the embryo, but not from the maternal tissues.
(3) It is composed of amniotic fluid.

A. (1) only
B. (1) and (2) only
C. (2) and (3) only
D. (1), (2) and (3)

##A##

Skill assessed	Mark
Understanding basic concepts	1

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|!|BCEMC0140000087|!|

The following processes are involved in the birth process in humans.

(1) The amnion breaks.
(2) Muscles of the uterine wall and the abdomen contract strongly to force the baby out.
(3) The placenta detaches from the uterine wall.
(4) The cervix dilates.

Which of the following is the correct order?

A. (1) → (4) → (3) → (2)
B. (3) → (4) → (1) → (2)
C. (4) → (1) → (2) → (3)
D. (4) → (1) → (3) → (2)

##C##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0140000088|!|

Which of the following contraceptive methods is the ***most*** reliable?

A. rhythm method
B. using condoms
C. taking contraceptive pills
D. sterilization

##D##

Skill assessed	Mark
Understanding basic concepts	1

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|!|BCEMC0140000089|!|

Which of the following statements about contraceptive pills is/are correct?

(1) They contain female sex hormones.
(2) They can prevent ovulation.
(3) Taking contraceptive pills is not a reliable contraceptive method.

A. (1) only
B. (1) and (2) only
C. (1) and (3) only
D. (2) and (3) only

##B##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0140000090|!|

Which of the following items can act as barrier(s) to prevent fertilization?

(1) condom
(2) diaphragm
(3) intrauterine device

A. (1) only
B. (3) only
C. (1) and (2) only
D. (2) and (3) only

##C##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0140000091|!|

What happens to a man if his sperm ducts are tied and cut?

A. He cannot produce sperms.
B. He cannot produce sex hormones.
C. His penis cannot become erect.
D. The semen ejaculated by him contains no sperms.

##D##

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Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0140000092|!|

Which of the following statements about the disadvantages of different contraceptive methods is ***incorrect***?

A. Taking hormonal contraceptives can cause side effects such as nausea and headache.
B. The rhythm method has a high failure rate.
C. An intrauterine device cannot be removed once it is put into the uterus.
D. Tubal ligation is usually irreversible.

##C##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0140000093|!|

Which of the following combinations ***incorrectly*** matches the contraceptive method with the biological basis behind?

	Contraceptive method	***Biological basis***
A.	using condoms	Sperms are prevented from meeting the ovum.
B.	tubal ligation	Ovulation is prevented.
C.	taking contraceptive pills	Ovulation is prevented.
D.	using intrauterine devices	Implantation of an embryo is prevented.

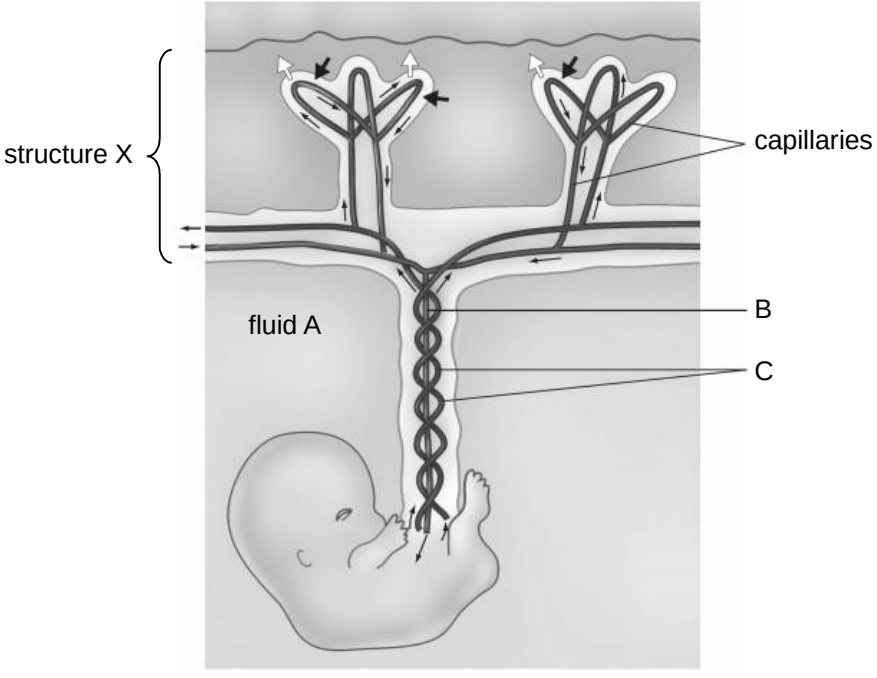
##B##

Skill assessed	Mark
Understanding basic concepts	1

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|!|BCEMB0140000094|!| ★

Directions: Questions (a) to (c) refer to the diagram below which shows a human embryo and the associated structures.



(a) Which of the following substances ***cannot*** pass from maternal blood to the embryo's blood through structure X?

A. nicotine
B. minerals
C. antibodies
D. plasma proteins

##D##

Skill assessed	Mark
Analysing data	1

|!|BCEMB0140000095|!| ★

(b) Which of the following is/are the function(s) of fluid A?

(1) to absorb shock
(2) to prevent desiccation of the embryo
(3) to carry away metabolic wastes from the embryo

A. (1) only
B. (1) and (2) only
C. (2) and (3) only
D. (1), (2) and (3)

##B##

Skill assessed	Mark
Analysing data	1

|!|BCEMB0140000096|!| ★

(c) Which of the following combinations about the compositions of blood in vessels B and C is correct?

	Blood in vessel B	*Blood in vessel C*
A.	more carbon dioxide	less carbon dioxide
B.	less glucose	more glucose
C.	less oxygen	more oxygen
D.	less urea	more urea

##D##

Skill assessed	Mark
Analysing data	1

|!|BCEMC0140000097|!|

Which of the following are the advantages of breast-feeding?

(1) Breast milk contains all the essential nutrients for the growth of baby.
(2) Breast milk is easy for babies to digest and absorb.
(3) Breast milk contains antibodies that can protect babies against pathogens.

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

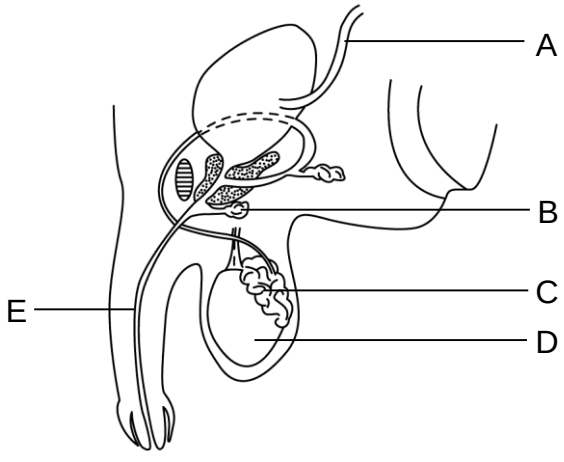
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##D##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0140000098|!|

Directions: Questions (a) to (c) refer to the diagram below which shows a part of the human male reproductive system.



(a) Which of the following combinations correctly identifies structures A and E?

	Structure A	***Structure E***
A.	ureter	sperm duct
B.	sperm duct	ureter
C.	ureter	urethra
D.	sperm duct	urethra

##C##

Skill assessed	Mark
Analysing data	1

|!|BCEMC0140000099|!|

(b) Which of the following structures is responsible for the production of sperms?

A. structure B
B. structure C
C. structure D
D. structure E

##C##

Skill assessed	Mark
Analysing data	1

|!|BCEMB0140000100|!| ★ **Cross topic**

(c) Which of the following plant structure has a similar function to structure D?

A. ovule
B. pollen tube
C. pollen grain
D. anther

##D##

Skill assessed	Mark
Analysing data	1

|!|BCEMC0140000101|!|

Which of the following contraception methods is the ***most*** effective in preventing the spread of HIV infection?

A. rhythm method
B. diaphragm
C. male condom
D. contraceptive pills

##C##

Skill assessed	Mark
Understanding basic concepts	1

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